



Greater Noida, Uttar Pradesh

INTERNSHIP PRESENTATION

Development of AI-Powered Automation Solutions for KDI Power

STUDENT NAME

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PROGRAM & SEMESTER

B.Tech CSE (AI/ML)

ENROLLMENT NUMBER

CS-2341412

ORGANIZATION

KDI Power Private Limited

OUTLINE

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INTRODUCTION

About KDI Power Private Limited

A New Delhi-based manufacturer of electrical wires and cables.

Manufactures LT Power Cables, XLPE, Aerial Bunched, Control Cables, House Wires and Rubber Cables.

Multiple information flows exist: customers need product info, sales teams need leads, technical teams work with datasheets.

Internship positioned within the organization's Artificial Intelligence and Automation activities.

SECTOR	Electrical Wire & Cable Mfg.
LOCATION	New Delhi, India
DOMAIN	Artificial Intelligence & Automation
ROLE	AI Operations Intern
INDUSTRY MENTOR	Mr. Vijay Gautam
WORKING MODEL	Hybrid Mon-Sat 10AM-6PM

GOALS

Internship Objectives

01

Identify Processes

Identify repetitive business processes suitable for automation.

02

Build AI Systems

Develop AI systems to understand messages, documents and data.

03

Automate Communication

Automate customer enquiry handling and lead capture via WhatsApp.

04

Lead Discovery

Build an automated B2B lead-discovery pipeline for contractors.

05

Competitive Intel

Automate competitive intelligence monitoring via LinkedIn.

06

Document Processing

Develop intelligent document-processing for scanned PDFs.

07

Task Automation

Automate internal task reminders via Sheets and WhatsApp.

08

Practical Experience

Gain experience in testing, deployment and maintenance.

CONTEXT

Areas Needing Automation

Five operational areas requiring substantial manual effort.

Customer enquiries received through messaging channels required direct human responses.

Lead discovery involved manually searching online sources and assembling company information.

Competitive monitoring required employees to inspect competitor and industry posts manually.

Scanned cable datasheets could not be conveniently edited using conventional PDF editors.

Internal operational tasks required follow-up communication between managers and employees.

Area 1**Customer communication & lead capture****Area 2****B2B lead discovery & qualification****Area 3****Competitive & tender intelligence****Area 4****Technical document processing****Area 5****Internal task tracking & reminders**

APPROACH

Development Methodology

PHASE 01

Requirement Discovery

Understood current business workflows and identified repetitive activities.

PHASE 02

Architecture & Selection

Selected technologies according to each problem (FastAPI, Playwright, etc.).

PHASE 03

Implementation

Systems developed incrementally: backend, AI integration, persistence, interfaces.

PHASE 04

Testing & Refinement

Webhook verification, incomplete inputs, API rate limits, Render cold starts.

PHASE 05

Deployment & Handover

Cloud deployment via Render & Supabase. Scheduled GitHub Actions.

PHASE 06

Documentation

Comprehensive documentation including architecture diagrams and setup guides.

FIGURE 4.1

Overall System Architecture

Key Workflow Steps

Business Requirements translated into specialized automation applications.

Applications use AI models, APIs, and data stores.

System produces structured business outputs like SQL, Excel, Sheets, and PDFs.

Five interconnected modules covering different operational areas.

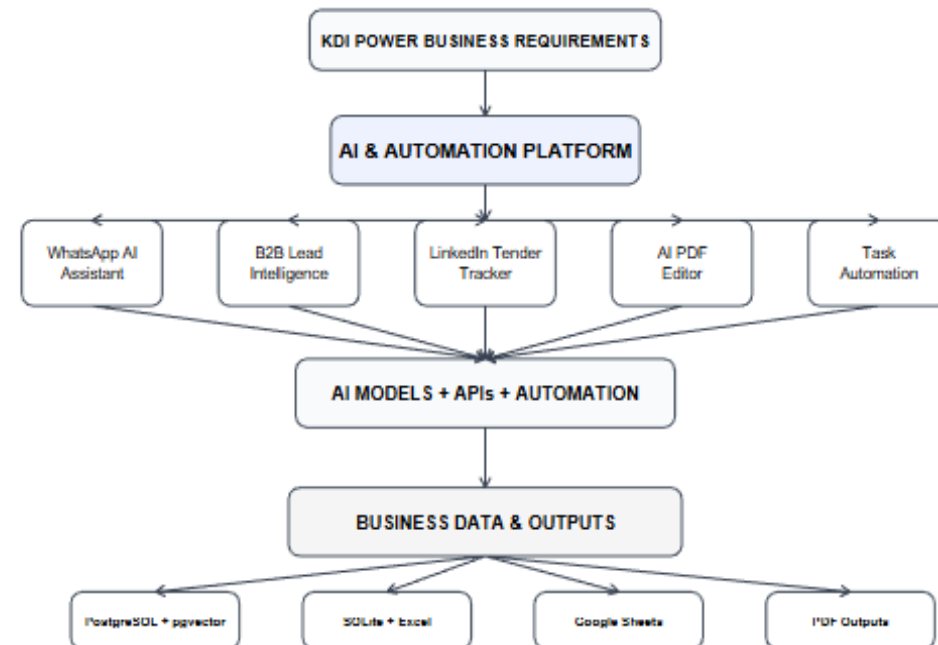


FIGURE 4.2

KDI Power AI WhatsApp Assistant

Key Workflow Steps

Customer sends WhatsApp message via Meta Cloud API.

Message forwarded to FastAPI webhook.

Groq LLM + RAG processes conversation with product context.

Lead extraction with structured tags ([LEAD_SUBMIT]).

Data stored in Supabase PostgreSQL.

Sales Dashboard visualizes leads and chat analytics.

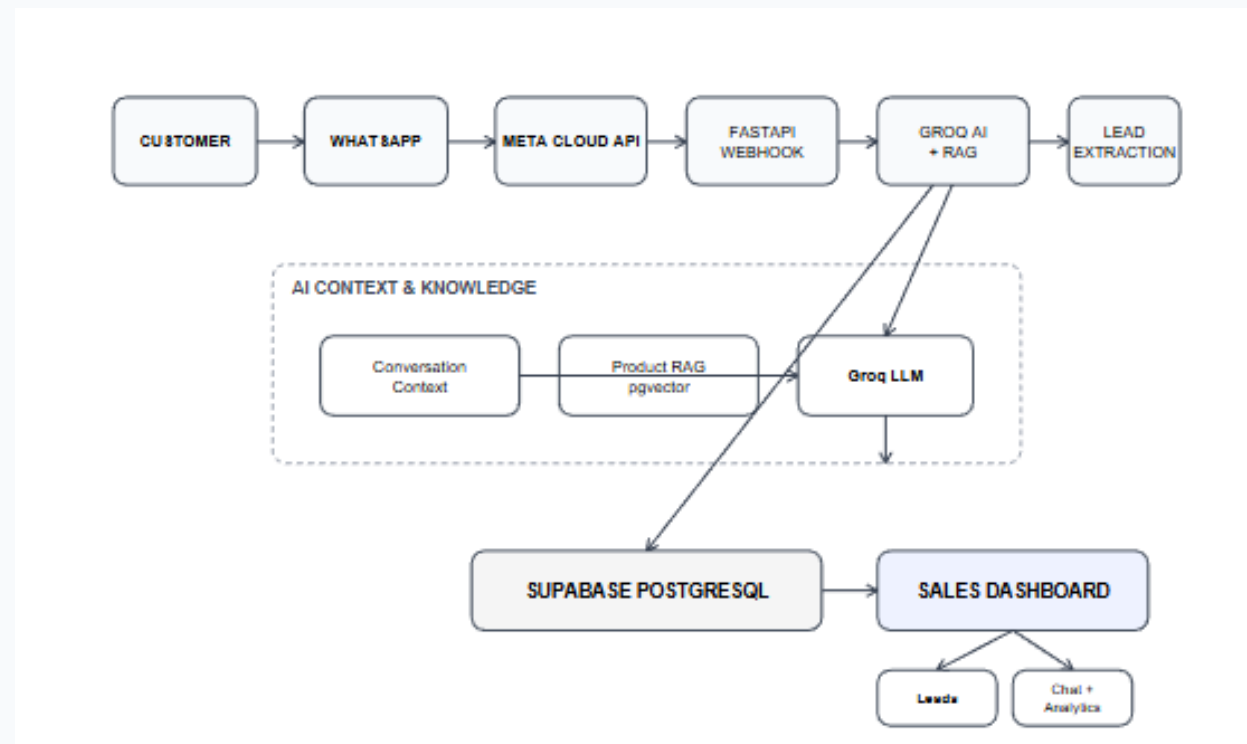


FIGURE 4.3

KDI Lead Intelligence & Finder

Key Workflow Steps

Tier 1 - Discovery: Playwright + Google Maps automation with proxy rotation.

Tier 2 - Intelligence: Website crawling, email validation, cable/sector signals.

Tier 3 - Qualification: Rule prefilter + Groq LLM + 0-100 scoring.

Outputs: Approved (75-100), Pending (50-74), Rejected (<50).

Target markets: India & South Africa.

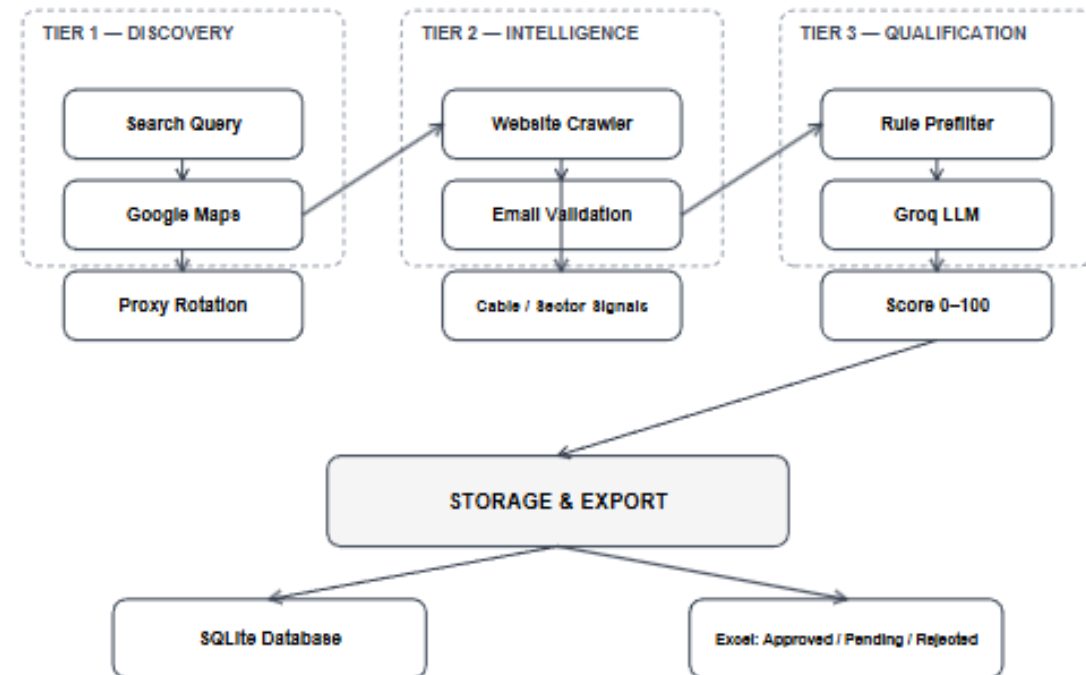


FIGURE 4.4

LinkedIn Tender Tracker

Key Workflow Steps

Playwright + CDP with persistent Chrome profile.

Relevance filter for tender, bid, RFP, cable, EPC, DISCOM keywords.

Mistral AI generates 3-5 bullet summaries.

Stored in Master_Tracker.xlsx.

Manual monitoring of 2-3 hours reduced to under 5 minutes.

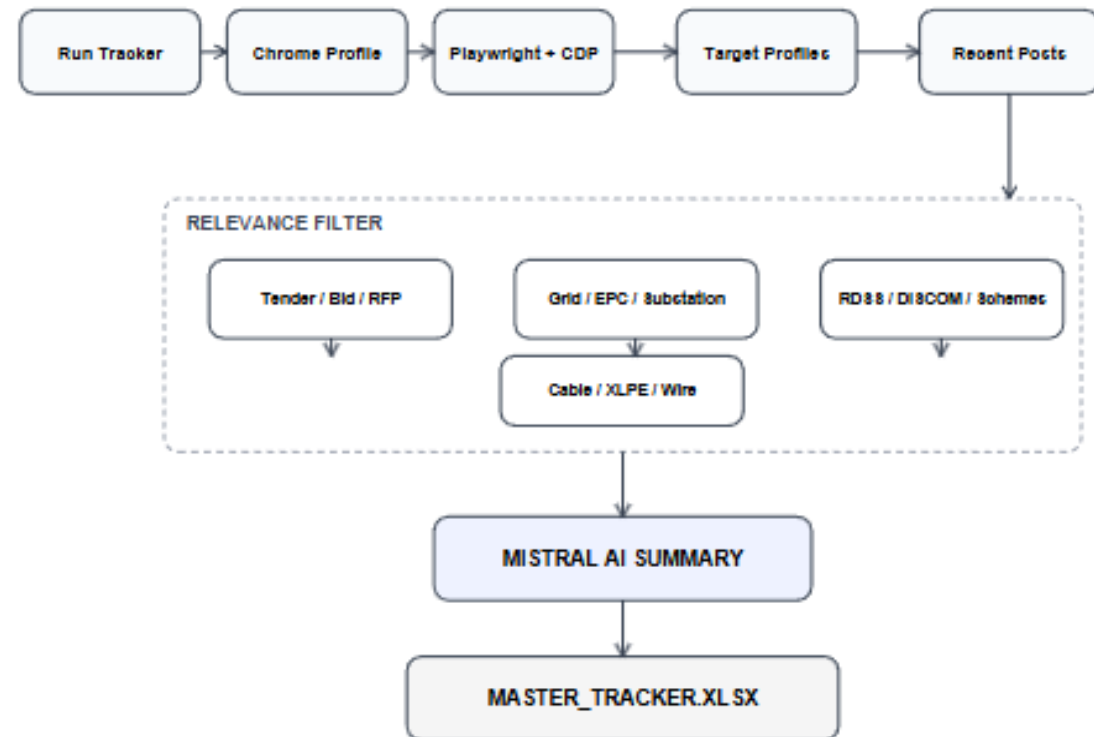


FIGURE 4.5

AI-Powered PDF Editor

Key Workflow Steps

MinerU for OCR and document layout analysis.

NVIDIA NIM Llama 3.2 90B Vision for tables & equations.

React 18 interface for interactive editing.

MathJax + Chromium for final PDF reconstruction.

Spec update reduced from 45-60 min to under 5 min.

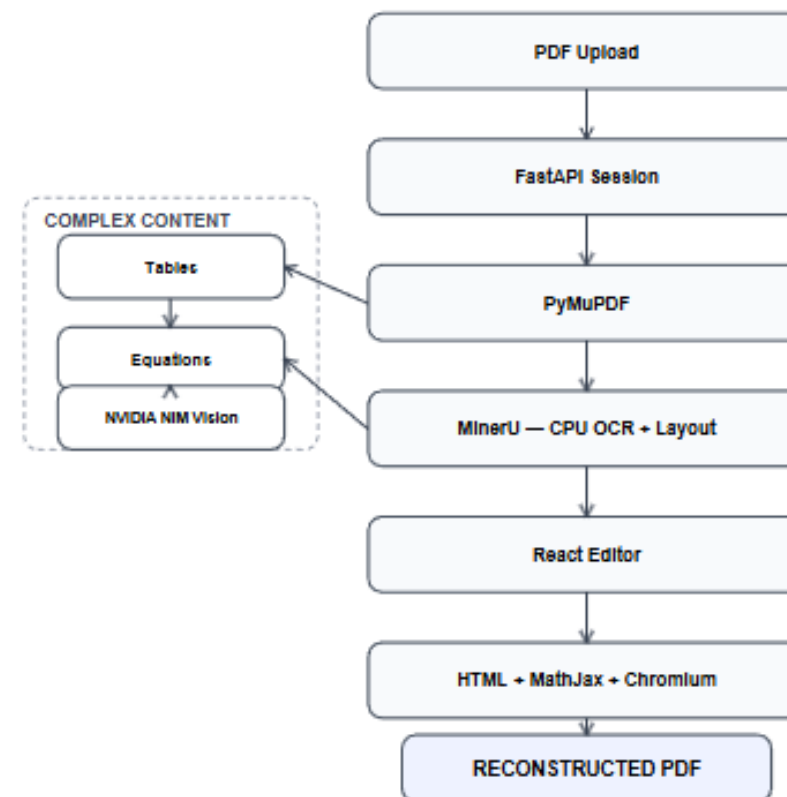


FIGURE 4.6

Task Automation Engine

Key Workflow Steps

Google Sheets API v4 connected to WhatsApp Cloud API.

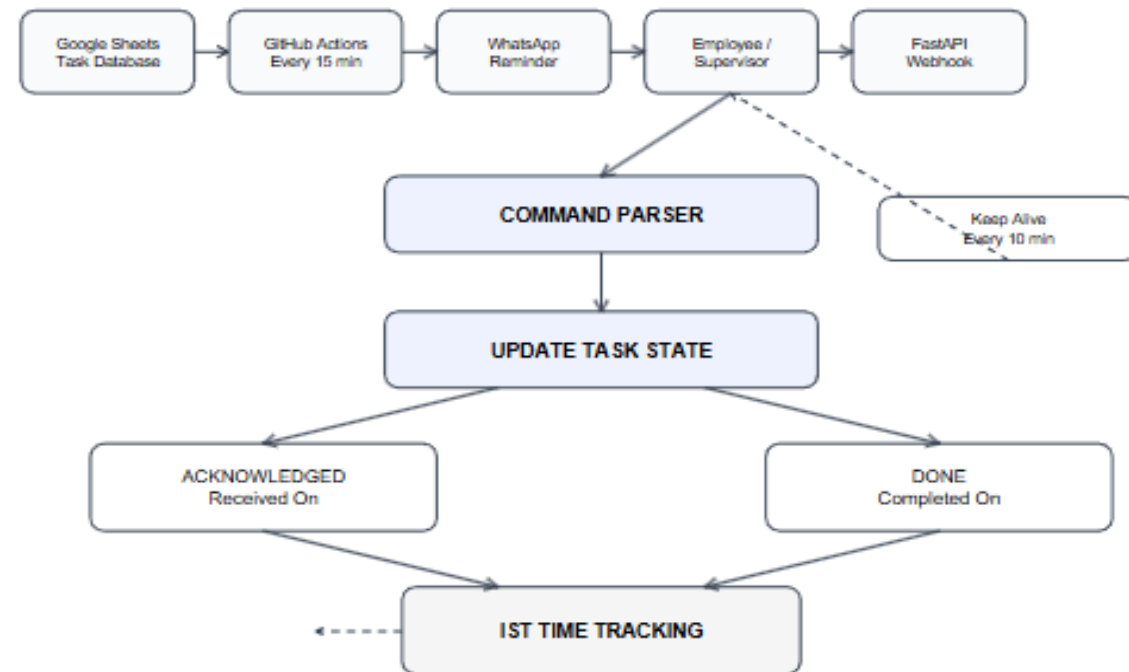
GitHub Actions runs reminder logic every 15 minutes.

Command parser handles 'ok' / 'done' replies.

States: Open -> Acknowledged -> Done.

IST timestamp tracking for received & completed times.

Keep-alive workflow every 10 minutes.



TOOLS

Technology Stack Used

Layer	Technology	Role
Programming	Python 3.10+	Core applications and automation scripts
Backend	FastAPI	APIs, webhooks and backend services
Frontend	React 18 + Vite, HTML/CSS/JS	Interactive PDF editor, Sales dashboard
Database	Supabase PostgreSQL, pgvector	Cloud lead, chat and product data (RAG)
AI / LLM	Groq, Mistral AI	Conversational AI, lead qualification, summaries
Vision AI	NVIDIA NIM Llama 3.2 90B Vision	Tables and mathematical formula extraction
Automation	Playwright, GitHub Actions	Google Maps/LinkedIn automation, scheduling
Cloud & APIs	Render, Meta WhatsApp API, Google Sheets	Deployment, messaging, task management

PROBLEM SOLVING

Technical Challenges & Approaches

CHALLENGES FACED

Third-party API rate limits (WhatsApp, Groq, Mistral, NVIDIA NIM).

Google Maps and LinkedIn bot detection during automation.

AI reliability - incomplete or unexpected model outputs.

CUDA memory constraints for vision-language processing.

Render cold starts affecting deployment responsiveness.

WhatsApp 24-hour service window & timezone differences.

Excel file locking during concurrent write operations.

APPROACHES TAKEN

Retries, fallbacks, approved templates and error handling.

Proxy rotation, persistent profiles, dynamic waiting.

Prompt engineering, validation, structured output formats.

CPU layout processing with cloud-based vision processing.

Keep-alive workflows, scheduled GitHub Actions.

Template messages and latest-open-task lookup logic.

Lock detection and retry logic for Excel operations.

OUTCOMES

Results, Impact & Learning

MEASURABLE EFFICIENCY GAINS

5

SYSTEMS

Operational areas addressed: customer inbound, B2B discovery, competitive monitoring, document processing, and task tracking.

< 5 mins

LINKEDIN MONITORING

Reduced from approximately 2-3 hours of manual monitoring per day to under five minutes for automated review.

< 5 mins

PDF UPDATE

Reduced from approximately 45-60 minutes of manual editing to under five minutes via the AI reconstruction pipeline.

KEY LEARNING OUTCOMES

End-to-End AI Engineering

Designed complete automation solutions from scratch - backend, AI integration, persistence, interfaces.

Production Discipline

Systems require structured I/O, error handling, fallbacks, API-limit handling, authentication, deployment.

Business Translation

Improved ability to translate business requirements into technical workflows that real users operate.

CONCLUSION

Thank You

For your time and attention throughout this presentation.

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KDI Power Pvt. Ltd.